

GREENING WATER USAGE



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Some pundits have suggested that the next international conflicts will not be over oil, but over water. Not long ago, an uproar was stirred in our area in the upper Midwest when it was discovered that a large ship from a foreign port was filling its hold (not just its ballast tanks) with water from one of the Great Lakes. (By the way, the Great Lakes make up one-fifth of the earth's surface fresh-water supply.)

Water has always been a trigger for tension when competing groups—such as farmers and ranchers, and now agricultural lands and cities—vie for the water they need for their survival. In the Southwest, one of the fastest-developing parts of our country, the primary aquifers are nearly depleted, and the Colorado River, which supplies most of the water to the region, has been drained to its limits. For those of us concerned with building performance, we cannot create structures that are sustainable without addressing water issues on both the interior and the exterior of these structures. Some new, imaginative strategies are emerging for this work, while some solutions are as old as civilization.

Once we recognize our responsibility to conserve water while planning and building structures, we have three basic tasks: to conserve water, to improve water quality, and to encourage efficient and speedy water heating and water distribution in-house. When we look at these tasks, we must also ask ourselves how to build effective water plans into our buildings and the lands surrounding them, and how to implement those plans.

Saving Water Inside...

Thanks to water conservation efforts over the past 15 years, we have made good headway on conserving water in our homes. When the 1.6 gallon per flush (gpf) toilet went into effect as the requirement for the Energy and Conservation Act of 1992, it was considered a controversial move. Many groups made huge efforts to lobby against it—first to keep it from being implemented and then to have the move reversed. The plumbing industry in general (manufacturers and installers) and many builders and contractors were opposed to the legislation and complained bitterly about the poor performance of the earliest versions of these toilets. However, as the quality and effectiveness of the product line has dramatically improved, opposition has nearly disappeared, and there is a general recognition that switching to the 1.6 gpf toilet was a good idea.

Now we should take the next step and make the use of toilets with double-flushing mechanisms mandatory. By switching to double-flush toilets, which use a half flush for liquid waste and a full flush for solid waste, a family of four with average daily flushes of six each can save up to 8,760 gallons of water each year. Considering that these toilets have been used in Australia for the last 25 years, it's obvious that those of us Stateside have some catching up to do.

You can now purchase bath faucets with a flow of 1.8 gallons per minute (gpm), kitchen faucets with a flow of 2 gpm, and showerheads with a flow of 2.5 gpm. As the marketplace is flooded with more low-flow showerheads that provide a shower that feels good, the public response to these efforts will improve.

To increase our water conservation within the home, I propose that in the near future, all water-using appliances should be Energy Star-rated. Graywater systems, which use tub, bath, washing



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machine, and sink drainage, as well as rainwater, to flush toilets and irrigate lawns, are installed on some homes, sometimes simply as rain barrels under eaves, and sometimes as more complex systems within the home (see "Graywater on the Grid," p. 12). However, the cost of proper installation may currently be unreasonable, given the amount of money and water that these systems save.

The point here is that we need to move full speed ahead on the most easily installed measures, such as low-flow devices, but carefully measure the costs and benefits of such measures as installing graywater systems, until we are sure these measures are affordable and appropriate in our individual residences. We can receive good counsel from thoughtful graywater advocates.

...and Out

Harvesting water with rain barrels is one of the simpler things we can do to conserve water outdoors. In almost every community, you can find rain

barrel workshops being offered through local conservation programs and adult schools. While most of these barrels do not hold enough water to make a huge dent in the water needs of the average yard, areas of the country with a serious water shortage are now using rain barrels containing 1,000 gallons or more (these are quite common in some parts of Europe).

One of the most interesting and oldest methods of retaining water on-site is the use of rain gardens. A rain garden is a small area in a landscape that is prepared to receive storm water from the roof or the yard. It includes plants and flowers that require little maintenance and that can survive in highly moist conditions. In cities where storm systems are overstressed and there is no money to expand or replace them, the use of rain gardens may significantly reduce water runoff and enhance the site at the same time.

The use of permeable surfaces, such as porous pavements, has emerged as another alternative for reducing surface runoff, but it is pricey and may not be a viable alternative at present. Recently, a major concrete supplier in Ohio sponsored a certification workshop for its clients on pervious concrete flatwork. Personally, I am sorry we gave up on gravel driveways. Maybe they'll be back!

The Water-Energy Connection

Water heating accounts for about 20% of average residential energy bills. How we heat water and transport the heated water to where it is used is a critical factor in helping reduce this energy cost. In all new construction, the water heater should be positioned so that the runs between the heater and the sites of use are as short as possible. I have taken advantage of this technique in my green rehab work in residential homes. While many building professionals are accustomed to seeing the hot water tank next to the furnace or boiler, putting the water heater closer to the kitchen and baths makes the most sense.

Many of the emerging green building guidelines from around the

country call for replacing hot water tanks with tankless water heaters. This eliminates the need to hold standing hot water in a tank—water that must be reheated several times before it is called for. The jury is still out as to which system is better, but the need for efficient, effective water heaters is clear.

In time, solar domestic hot water (DHW) will be cost-effective enough to include in every project. In the short term, we should include it in enough projects that we can learn more fully how to benefit from its use. We must improve our data-gathering abilities on water conservation work so we are sure we are going where we need to go, and are getting results that are consistent with our objectives. Insulating our hot water pipes (and our cold water pipes, to eliminate condensation and related moisture problems) is another measure that can reduce the energy cost of producing hot water.

Finally, we need to pay attention to the quality of our water. In the May 2007 issue of *Consumer Reports*, there is an excellent summary of the variety of water treatment measures that can be employed in our homes. There are pluses and minuses for each. The article reminds us that consumer confidence reports, which detail how the water coming to your home is being treated and if it is being treated effectively, can be obtained from your local water department. Persons living with wells in rural areas need to have their water tested, given the potential for contaminants entering a well-based water supply.

For years the water entry lines to our homes were made of lead, and the solder used to put copper piping together had lead in it. Over time, mineral deposits in the pipes reduced the amount of lead that leached into the water, and as our plumbing has been rehabbed, the threat of lead contamination has diminished—especially since 1988, when lead was banned from solder.

For us, as building performance personnel, the time has come to include greening water usage as part of our tool kit. While I have never had a

client who was building or renovating a house specifically for the purpose of conserving water, this has often been a secondary priority for homeowners I've worked with. For instance, in new construction, our first green home used a hot water tank as the primary heat source, so the whole house benefited from the hot water tank, even when it was not heating water for bathing, cooking, or washing dishes or clothes. I still like this combo system in a not-too-large house, as it eliminates the need for two primary heating sources in a home.

Water use in homes is becoming an important issue, and many homeowners are addressing it in one way or another. Most of my clients have focused on one or two water conservation or water quality issues while remodeling or building new. Some have focused on water purifiers (especially the reverse-osmosis version, though the downside of such units is that it takes several gallons of water to get a gallon of processed water). Many of my clients have insulated their water pipes, because that was something they could do themselves. I have also been surprised at the number of persons signing up for rain barrel workshops.

There was recently an article in the *Cleveland Plain Dealer* about rain gardens. This type of broad-interest coverage usually means that the subject has finally made it onto the general public's radar screen. After more such articles are published in local newspapers I am convinced that there will be great interest in rain gardens among the general populace, because it is far cheaper to install a rain garden than it is to excavate around foundations to replace old storm drains that are leaking through basement walls.



For more information:

To learn more about graywater, go to www.oasisdesign.net/greywater/misinfo/index.htm.